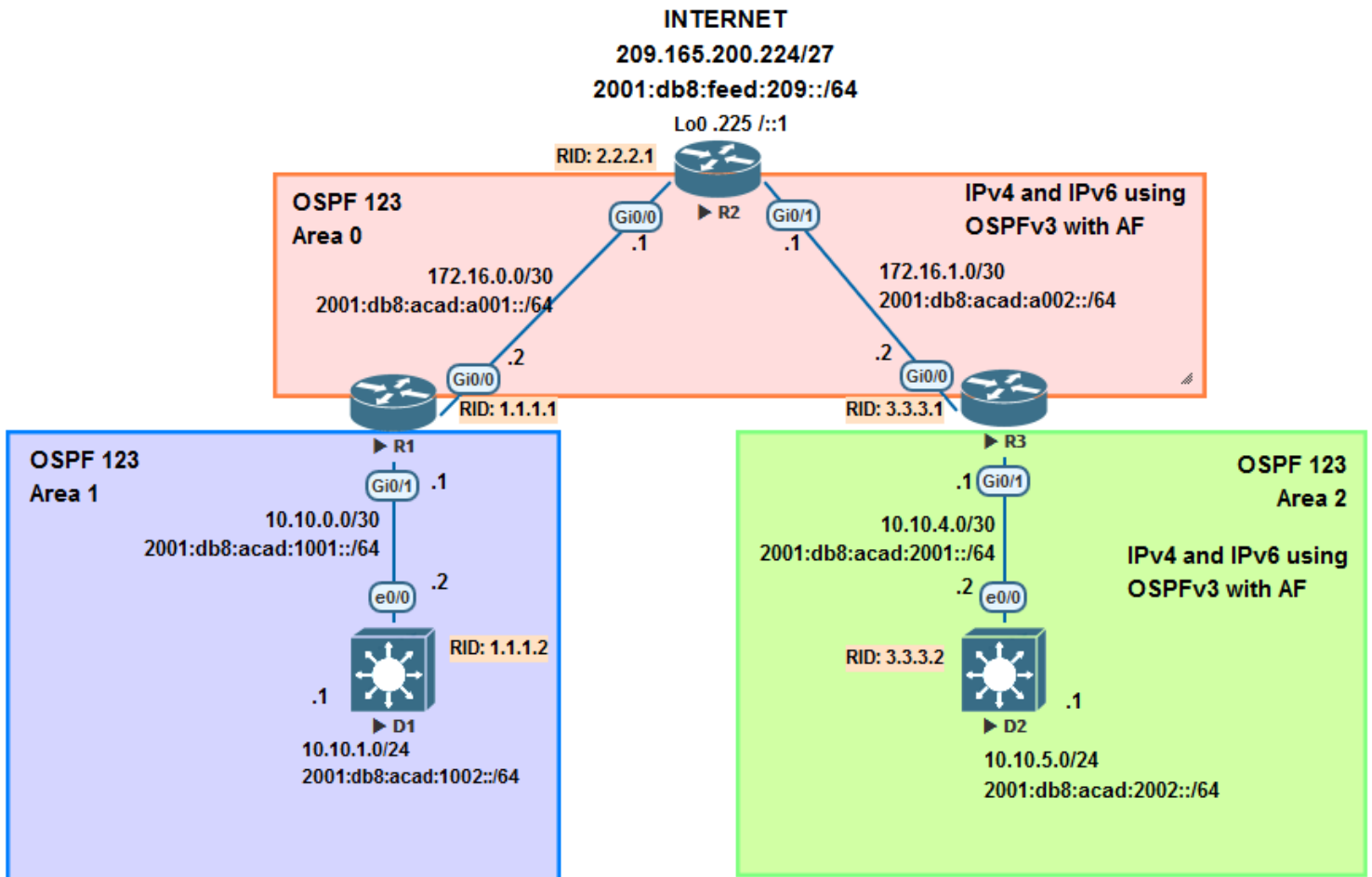


Lab - Implement Multiarea OSPFv3 Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
R1	G0/0	172.16.0.2/30	2001:db8:acad:a001::2/64	fe80::1:2
	G0/1	10.10.0.1/30	2001:db8:acad:1001::1/64	fe80::1:1
R2	Lo0	209.165.200.225/27	2001:db8:feed:209::1/64	fe80::2:3
	G0/0	172.16.0.1/30	2001:db8:acad:a001::1/64	fe80::2:1
	G0/1	172.16.1.1/30	2001:db8:acad:a002::1/64	fe80::2:2
R3	G0/0	172.16.1.2/30	2001:db8:acad:a002::2/64	fe80::3:2
	G0/1	10.10.4.1/30	2001:db8:acad:2001::1/64	fe80::3:1
D1	G0/0	10.10.0.2/30	2001:db8:acad:1001::2/64	fe80::d1:2
	G0/1	10.10.1.0/24	2001:db8:acad:1002::1/64	fe80::d1:1

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
D2	G0/0	10.10.4.2/30	2001:db8:acad:2001::2/64	fe80::d2:2
	G0/1	10.10.5.1/24	2001:db8:acad:2002::1/64	fe80::d2:1

Objectives

Part 1: Build the Topology and Configure Basic Device Settings and IP Addressing

Part 2: Configure Traditional OSPFv3 for IPv6 on D1

Part 3: Configure OSPFv3 for Address Families (AF) IPv4 and AF IPv6

Part 4: Verify OSPFv3 AF

Part 5: Tune OSPFv3 AF

Background / Scenario

In this lab, you will configure the network with multiarea OSPFv3 routing using the AF feature for both IPv4 and IPv6 in OSPF areas 0, 1 and 2. This lab was specifically designed to use three routers and two Layer 3 switches that support OSPFv3 using AF.

It should be noted that OSPFv3 runs on top of IPv6 and uses IPv6 link local addresses for OSPFv3 control packets. Therefore, it is required that IPv6 be enabled on an OSPFv3 link, although the link may not be participating in any IPv6 AFs. Additionally, OSPFv3 AF for IPv4 unicast is not backwards compatible with OSPFv2.

Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers and switches.

Step 1: Configure basic settings for each router.

- Console into each device, enter global configuration mode, and apply the basic settings and interface addressing using the following startup configurations for each device.

Router R1

```
hostname R1
no ip domain lookup
line con 0
  logging sync
  exec-timeout 0 0
exit
interface G0/0
  ip add 172.16.0.2 255.255.255.252
  ipv6 add 2001:db8:acad:a001::2/64
  ipv6 add fe80::1:2 link-local
  no shut
exit
interface Ethernet0/1
  ip add 10.10.0.1 255.255.255.0
  ipv6 add 2001:db8:acad:1001::1/64
```

```
ipv6 add fe80::1:1 link-local
no shut
exit
```

Router R2

```
hostname R2
no ip domain lookup
line con 0
  logging sync
  exec-time 0 0
exit
interface G0/0
  ip add 172.16.0.1 255.255.255.252
  ipv6 add 2001:db8:acad:a001::1/64
  ipv6 add fe80::2:1 link-local
  no shut
  exit
interface Ethernet0/1
  ip address 172.16.1.1 255.255.255.252
  ipv6 add 2001:db8:acad:a002::1/64
  ipv6 add fe80::2:2 link-local
  no shut
  exit
int lo0
  ip add 209.165.200.225 255.255.255.224
  ipv6 add 2001:db8:feed:209::1/64
  ipv6 add fe80::2:3 link-local
  exit
```

Router R3

```
hostname R3
no ip domain lookup
line con 0
  logging sync
  exec-time 0 0
exit
interface G0/0
  ip add 172.16.1.2 255.255.255.252
  ipv6 add 2001:db8:acad:a002::2/64
  ipv6 add fe80::3:2 link-local
  no shut
  exit
interface Ethernet0/1
  ip address 10.10.4.1 255.255.255.252
  ipv6 add 2001:db8:acad:2001::1/64
  ipv6 add fe80::3:1 link-local
```

```
no shut
exit
```

Switch D1

```
hostname D1
no ip domain lookup
line con 0
  exec-timeout 0 0
  logging synchronous
exit
interface e0/0
  no switchport
  ip add 10.10.0.2 255.255.255.0
  ipv6 add 2001:db8:acad:1001::2/64
  ipv6 add fe80::d1:2 link-local
  no shutdown
exit
interface e0/1
  no switchport
  ip add 10.10.1.1 255.255.255.0
  ipv6 add 2001:db8:acad:1002::1/64
  ipv6 add fe80::d1:1 link-local
  no shutdown
exit
```

Switch D2

```
host D2
no ip domain lookup
line con 0
  logging sync
  exec-time 0 0
exit
interface e0/0
  no switchport
  ip address 10.10.4.2 255.255.255.252
  ipv6 add 2001:db8:acad:2001::2/64
  ipv6 add fe80::d2:2 link-local
  no shut
exit
interface e0/1
  no switchport
  ip address 10.10.5.1 255.255.255.0
  ipv6 add 2001:db8:acad:2002::1/64
  ipv6 add fe80::d2:1 link-local
  no shut
exit
```

- b. Save the running configuration to startup-config.

Part 2: Configure Traditional OSPFv3 for IPv6 on D1

Step 1: Configure traditional OSPFv3 on D1.

Traditional OSPFv3 implements OSPF routing for IPv6. In this part of the lab, you will configure traditional OSPFv3 for routing IPv6 on D1, which is in the IPv6-only area.

- a. OSPFv3 messages are sourced from the router's IPv6 link-local address. Earlier in this lab, IPv6 GUA and link-local addresses were statically configured on each router's interface. The link-local addresses were statically configured to make these addresses more recognizable than being automatically created using EUI-64. Issue the **show ipv6 interface brief** command to verify the GUA and link-local addresses on the router's interfaces.

```
D1# show ipv6 interface brief
```

```
Ethernet0/0          [up/up]
    FE80::D1:2
    2001:DB8:ACAD:1001::2
Ethernet0/1          [up/up]
    FE80::D1:1
    2001:DB8:ACAD:1002::1
Ethernet0/2          [up/up]
    unassigned
Ethernet0/3          [up/up]
    unassigned
```

- b. IPv6 routing is disabled by default. Enable IPv6 routing using the **ipv6 unicast-routing** command in global configuration mode.

```
D1(config)# ipv6 unicast-routing
```

- c. Most Cisco IOS versions have IPv6 CEF enabled by default when IPv6 routing is enabled. Use the **show ipv6 cef** command to verify whether IPv6 CEF is enabled. If you need to enable IPv6 CEF, use the **ipv6 cef** command. If IPv6 CEF is disabled you will see the an IOS message similar to "%IPv6 CEF not running".

```
D1# show ipv6 cef
```

```
::/0
  no route
::/127
  discard
2001:DB8:ACAD:1001::/64
  attached to Ethernet0/0
2001:DB8:ACAD:1001::2/128
  receive for Ethernet0/0
2001:DB8:ACAD:1002::/64
  attached to Ethernet0/1
2001:DB8:ACAD:1002::1/128
  receive for Ethernet0/1
FE80::/10
  receive for Null0
FF00::/8
  multicast
FF02::/16
```

receive

- d. Configure the OSPFv3 process on D1. Similar to OSPFv2, the process ID does not have to match other routers to form neighbor adjacencies, although that is considered best practice. Configure the 32-bit OSPFv3 router ID on each router. Enable OSPFv3 directly on the interfaces using the interface **ipv6 ospf pid area area** command.

```
D1(config)# ipv6 unicast-routing
D1(config)# ipv6 router ospf 123
D1(config-rtr)# router-id 1.1.1.2
D1(config-rtr)# exit
D1(config)# interface e0/0
D1(config-if)# ipv6 ospf 123 area 1
D1(config-if)# exit
D1(config)# interface e0/1
D1(config-if)# ipv6 ospf 123 area 1
D1(config-if)# exit
```

- e. The **show ipv6 ospf** command can be used to verify the OSPF router ID. If the OSPFv3 router ID is uses a 32-bit value other than the one specified by the **router-id** command, you can reset the router ID by using the **clear ipv6 ospf pid process** command and re-verify using the command **show ipv6 ospf**.

```
D1# show ipv6 ospf
```

```
Routing Process "ospfv3 123" with ID 1.1.1.2
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msec
Minimum hold time between two consecutive SPF 200 msec
Maximum wait time between two consecutive SPF 5000 msec
Initial LSA throttle delay 50 msec
Minimum hold time for LSA throttle 200 msec
Maximum wait time for LSA throttle 5000 msec
Minimum LSA arrival 100 msec
LSA group pacing timer 240 sec
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
Retransmission limit dc 24 non-dc 24
EXCHANGE/LOADING adjacency limit: initial 300, process maximum 300
Number of external LSA 0. Checksum Sum 0x000000
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Graceful restart helper support enabled
Reference bandwidth unit is 100 mbps
RFC1583 compatibility enabled
```

```
Area 1
```

```
Number of interfaces in this area is 2
SPF algorithm executed 5 times
Number of LSA 12. Checksum Sum 0x0486C1
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
```

Flood list length 0

- f. The **show ipv6 protocols** command can be used to verify general OSPFv3 information such as areas and enabled interfaces.

```
D1# show ipv6 protocols
IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 123"
  Router ID 1.1.1.2
  Number of areas: 1 normal, 0 stub, 0 nssa
  Interfaces (Area 1):
    Ethernet0/1
    Ethernet0/0
  Redistribution:
    None
```

Part 3: Configure OSPFv3 for AF IPv4 and AF IPv6

OSPFv3 with the address family (AF) unifies OSPF configuration for both IPv4 and IPv6. Each OSPFv3 AF is a single process, so you may have two processes per interface, but only one process per AF. OSPFv3 messages are sent over IPv6 which requires that IPv6 routing is enabled and that the interface has a link-local IPv6 address. This is the requirement even if only the IPv4 AF is configured.

In this section you will configure OSPFv3 with AF for the IPv4 and IPv6 address families on R1, R2, R3, D1 and D2.

Step 1: Configure OSPFv3 with AF on R1.

- a. After enabling IPv6 unicast routing, configure OSPFv3 with AF on R1 using the **router ospfv3 pid** command. Use the **?** to see the address families available.

```
R1(config)# ipv6 unicast-routing
R1(config)# router ospfv3 123
R1(config-router)# address-family ?
  ipv4  Address family
  ipv6  Address family
```

- b. Next, specify the AF for IPv4 and use the **?** to see the available options.

```
R1(config-router)# address-family ipv4 ?
  unicast  Address Family modifier
  vrf      Specify parameters for a VPN Routing/Forwarding instance
  <cr>
```

- c. Enter the AF for IPv4 unicast using the command **address-family ipv4 unicast**. Use the **?** to examine the options in AF configuration mode. Some of the more common configuration commands are highlighted. Use the **router-id** command to configure the router ID for the IPv4 AF.

```
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)# ?
Router Address Family configuration commands:
  adjacency          Control adjacency formation
  area                OSPF area parameters
  authentication      Authentication parameters
  auto-cost           Calculate OSPF interface cost according to bandwidth
  auto-cost-determination Calculate OSPF interface cost according to bandwidth
```

bfd	BFD configuration commands
compatible	Compatibility list
default	Set a command to its defaults
default-information	Control distribution of default information
default-metric	Set metric of redistributed routes
discard-route	Enable or disable discard-route installation
distance	Define an administrative distance
distribute-list	Filter networks in routing updates
event-log	Event Logging
exit-address-family	Exit from Address Family configuration mode
graceful-restart	Graceful-restart options
help	Description of the interactive help system
ignore	Do not complain about specific event
interface-id	Source of the interface ID
limit	Limit a specific OSPF feature
local-rib-criteria	Enable or disable usage of local RIB as route criteria
log-adjacency-changes	Log changes in adjacency state
manet	Specify MANET OSPF parameters
max-lsa	Maximum number of non self-generated LSAs to accept
max-metric	Set maximum metric
maximum-paths	Forward packets over multiple paths
mpls	MPLS Traffic Engineering configs
no	Negate a command or set its defaults
passive-interface	Suppress routing updates on an interface
prefix-suppression	Enable prefix suppression
process-min-time	Percentage of quantum to be used before releasing CPU
queue-depth	Hello/Router process queue depth
redistribute	Redistribute information from another routing protocol
router-id	router-id for this OSPF process
shutdown	Shutdown the router process
snmp	Modify snmp parameters
statistics	Enable or disable OSPF statistics options
summary-address	Configure IP address summaries
summary-prefix	Configure IP address summaries
timers	Adjust routing timers

```
R1(config-router-af) #
```

```
R1(config-router-af) # router-id 1.1.1.1
```

- d. Exit the IPv4 AF configuration mode and enter the AF IPv6 configuration mode. The **exit-address-family** (or a shorter version of **exit**) command is used exit address family configuration mode. Issue the **address-family ipv6 unicast** command to enter the IPv6 AF. For the IPv6 AF, use the **router-id** command to configure the router ID. It isn't necessary to configure a different router ID for IPv6 AF but it is a valid option. The **exit** command is used to return to global configuration mode.

```
R1(config-router-af) # exit-address-family
```

```
R1(config-router) # address-family ipv6 unicast
```

```
R1(config-router-af) # router-id 1.1.1.1
```

```
R1(config-router-af) # exit-address-family
```



```
R1(config-router)# exit
```

- e. OSPFv3 is enabled directly on the interfaces for both IPv4 and IPv6 AFs using the **ospfv3 pid [ipv4 | ipv6] area area-id** interface command. Use this command to enable OSPFv3 on both of R1's interfaces.

```
R1(config)# interface G0/0
R1(config-if)# ospfv3 123 ipv4 area 0
R1(config-if)# ospfv3 123 ipv6 area 0
R1(config-if)# exit
R1(config)# interface G0/1
R1(config-if)# ospfv3 123 ipv4 area 1
R1(config-if)# ospfv3 123 ipv6 area 1
```

Step 2: Configure OSPFv3 with AF IPv4 and AF IPv6 on R2.

Enable IPv6 unicast routing and configure the OSPFv3 with AF for both IPv4 and IPv6 on R2, similar to the configuration for R1.

```
R2(config)# ipv6 unicast-routing

R2(config)# router ospfv3 123
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)# router-id 2.2.2.1
R2(config-router-af)# exit-address-family
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)# router-id 2.2.2.1
R2(config-router-af)# exit-address-family
R2(config-router)# exit

R2(config)# interface G0/0
R2(config-if)# ospfv3 123 ipv4 area 0
R2(config-if)# ospfv3 123 ipv6 area 0
R2(config-if)# exit
R2(config)# interface G0/1
R2(config-if)# ospfv3 123 ipv4 area 0
R2(config-if)# ospfv3 123 ipv6 area 0
```

Step 3: Configure OSPFv3 with IPv4 AF and IPv6 AF on R3.

Enable IPv6 unicast routing and configure the OSPFv3 with AF for both IPv4 and IPv6 on R3, similar to the configurations for R1 and R2. On R3, set the router ID for both IPv4 AF and IPv6 AF with a single command as shown.

```
R3(config)# ipv6 unicast-routing

R3(config)# router ospfv3 123
R3(config-router)# router-id 3.3.3.1
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)# exit-address-family
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)# exit-address-family
```

```
R3(config-router)# exit

R3(config)# interface G0/0
R3(config-if)# ospfv3 123 ipv4 area 0
R3(config-if)# ospfv3 123 ipv6 area 0
R3(config-if)# exit
R3(config)# interface G0/1
R3(config-if)# ospfv3 123 ipv4 area 2
R3(config-if)# ospfv3 123 ipv6 area 2
```

Step 4: Configure OSPFv3 with AF on D2.

- a. Enter the following command to enable routing for IPv4. (This may not be required on depending on model and IOS.)

```
D2(config)# ip routing
```

- b. Enter the following command to enable routing for IPv6. (This may not be required on depending on model and IOS.)

```
D2(config)# ipv6 unicast-routing
```

Note: By default, the 3650 supports IPv6 interface configuration.

- c. Configure the OSPFv3 with AF for both IPv4 and IPv6 on D2, similar to the configurations for R1, R2 and R3.

```
D2(config)# router ospfv3 123
D2(config-router)# address-family ipv4 unicast
D2(config-router-af)# router-id 3.3.3.2
D2(config-router-af)# exit-address-family
D2(config-router)# address-family ipv6 unicast
D2(config-router-af)# router-id 3.3.3.2
D2(config-router-af)# exit-address-family
D2(config-router)# exit
```

```
D2(config)# interface e0/0
D2(config-if)# ospfv3 123 ipv4 area 2
D2(config-if)# ospfv3 123 ipv6 area 2
D2(config-if)# exit
D2(config)# interface e0/1
D2(config-if)# ospfv3 123 ipv4 area 2
D2(config-if)# ospfv3 123 ipv6 area 2
```

Part 4: Verify OSPFv3

The commands to verify traditional OSPFv3 and OSPFv3 with AF may differ. This is because OSPFv3 with AF commands include information for both IPv4 and IPv6 address families, whereas traditional OSPFv3 is for IPv6 only.

Step 1: Verifying neighbor adjacencies.

- a. Use the **show ipv6 ospf neighbor** command on D1 to display OSPFv3 neighbors. This is a command used for routers configured with traditional OSPFv3. The equivalent command for OSPFv2 would be **show ip ospf neighbor**.

D1# **show ipv6 ospf neighbor**

OSPFv3 Router with ID (1.1.1.2) (Process ID 123)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	1	FULL/DR	00:00:39	6	Ethernet0/0

- b. This same command on a router running OSPFv3 with AF would generate similar output. For example, on R1 issue the same **show ipv6 ospf neighbor** command. Notice the output is only OSPFv3 for the IPv6 AF.

R1# **show ipv6 ospf neighbor**

OSPFv3 Router with ID (1.1.1.1) (Process ID 123)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
2.2.2.1	1	FULL/BDR	00:00:31	5	GigabitEthernet0/0
1.1.1.2	1	FULL/BDR	00:00:38	471	GigabitEthernet0/1

- c. Now, issue the **show ospfv3 neighbor** command on R1. This is a command used for routers configured for OSPFv3 with AF. Notice the output includes neighbors for both IPv4 and IPv6 address families.

R1# **show ospfv3 neighbor**

OSPFv3 123 address-family ipv4 (router-id 1.1.1.1)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
2.2.2.1	1	FULL/BDR	00:00:38	5	GigabitEthernet0/0

OSPFv3 123 address-family ipv6 (router-id 1.1.1.1)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
2.2.2.1	1	FULL/BDR	00:00:32	5	GigabitEthernet0/0
1.1.1.2	1	FULL/BDR	00:00:30	471	GigabitEthernet0/1

Traditional OSPFv3 commands are similar to those for OSPFv2, except **ipv6** is used as an argument instead of **ip**, for example **show ip ospf neighbor** and **show ipv6 ospf neighbor**. OSPFv3 with AF uses the argument **ospfv3** which includes both OSPF for IPv4 and IPv6 AFs. For example, **show ospfv3 neighbor**.

Traditional OSPFv3 commands can be used when a router is configured for OSPFv3 with AF, but the OSPFv3 AF router will only show OSPF for IPv6 AF information. OSPFv3 with AF commands cannot be used on routers configured with traditional OSPFv3.

To summarize the **show** command arguments:

- OSPFv2: Use **show ip ospf** (IPv4 only)
- Traditional OSPFv3: Use **show ipv6 ospf** (IPv6 only)
- OSPFv3 with AF: Use **show ospfv3** (IPv4 and IPv6 AF) or **show ipv6 ospf** (IPv6 only)

Step 2: Examining the IP routing tables.

- Use the **show ipv6 route ospf** command on D1 to display OSPFv3 routing entries in the IPv6 routing table.

```
D1# show ipv6 route ospf
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, la - LISP away, le - LISP extranet-policy
OI 2001:DB8:ACAD:2001::/64 [110/13]
   via FE80::1:1, Ethernet0/0
OI 2001:DB8:ACAD:2002::/64 [110/23]
   via FE80::1:1, Ethernet0/0
OI 2001:DB8:ACAD:A001::/64 [110/11]
   via FE80::1:1, Ethernet0/0
OI 2001:DB8:ACAD:A002::/64 [110/12]
   via FE80::1:1, Ethernet0/0
```

Display the routes using the **show ip route ospf**. Why are there no routes displayed using this command?

- Understanding the difference between commands associated with OSPFv2 and OSPFv3 can seem challenging at times. The **show ip route ospfv3** command is used to view OSPFv3 routes in the IPv4 routing table. The **show ipv6 route ospf** command is used to view OSPFv3 routes in the IPv6 routing table. The **show ipv6 route ospf** command is the same command used with traditional OSPFv3 for IPv6.

```
R1# show ip route ospf

R1# show ip route ospfv3
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
O IA   10.10.4.0/30 [110/3] via 172.16.0.1, 00:04:47, GigabitEthernet0/0
O IA   10.10.5.0/24 [110/13] via 172.16.0.1, 00:04:47, GigabitEthernet0/0
    172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O      172.16.1.0/30 [110/2] via 172.16.0.1, 00:04:47, GigabitEthernet0/0
```

```
R1# show ipv6 route ospfv3
```

^

```
% Invalid input detected at '^' marker.
```

```
R1# show ipv6 route ospf
```

```
IPv6 Routing Table - default - 9 entries
```

```
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
```

```
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
```

```
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
```

```
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
```

```
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
```

```
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
```

```
       ON2 - OSPF NSSA ext 2, a - Application
```

```
O   2001:DB8:ACAD:1002::/64 [110/11]  
    via FE80::D1:2, GigabitEthernet0/1
```

```
OI  2001:DB8:ACAD:2001::/64 [110/3]  
    via FE80::2:1, GigabitEthernet0/0
```

```
OI  2001:DB8:ACAD:2002::/64 [110/13]  
    via FE80::2:1, GigabitEthernet0/0
```

```
O   2001:DB8:ACAD:A002::/64 [110/2]  
    via FE80::2:1, GigabitEthernet0/0
```

Why doesn't the **show ip route ospf** command display any routes on R1?

Step 3: Examining the OSPF LSDB.

- D1 is running traditional OSPFv3. The **show ipv6 ospf database** command is used to display a summary of the OSPFv3 LSDB.

```
D1# show ipv6 ospf database
```

```
OSPFv3 Router with ID (1.1.1.2) (Process ID 123)
```

```
Router Link States (Area 1)
```

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
1.1.1.1	1096	0x80000009	0	1	B
1.1.1.2	1110	0x80000005	0	1	None

```
Net Link States (Area 1)
```

ADV Router	Age	Seq#	Link ID	Rtr count
1.1.1.1	1152	0x80000001	6	2

```
Inter Area Prefix Link States (Area 1)
```

ADV Router	Age	Seq#	Prefix
1.1.1.1	1096	0x80000003	2001:DB8:ACAD:A001::/64
1.1.1.1	1096	0x80000003	2001:DB8:ACAD:A002::/64
1.1.1.1	833	0x80000005	2001:DB8:ACAD:2001::/64
1.1.1.1	1497	0x80000002	2001:DB8:ACAD:2002::/64

Link (Type-8) Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Interface
1.1.1.2	1150	0x80000001	39	G0/1
1.1.1.1	1096	0x80000006	6	G0/0
1.1.1.2	1151	0x80000001	38	G0/0

Intra Area Prefix Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Ref-lsttype	Ref-LSID
1.1.1.1	1152	0x80000001	6144	0x2002	6
1.1.1.2	1150	0x80000003	0	0x2001	0

- b. R1 is running OSPFv3 with AF. The **show ospfv3 database** command is used to display a summary of the OSPFv3 LSDB for both the IPv4 and IPv6 AFs.

R1# **show ospfv3 database**

OSPFv3 123 address-family ipv4 (router-id 1.1.1.1)

Router Link States (Area 0)

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
1.1.1.1	532	0x80000005	0	1	None
2.2.2.1	508	0x80000008	0	2	None
3.3.3.1	507	0x80000006	0	1	B

Net Link States (Area 0)

ADV Router	Age	Seq#	Link ID	Rtr count
2.2.2.1	539	0x80000001	5	2
3.3.3.1	512	0x80000001	5	2

Inter Area Prefix Link States (Area 0)

ADV Router	Age	Seq#	Prefix
3.3.3.1	553	0x80000001	10.10.4.0/30
3.3.3.1	513	0x80000001	10.10.5.0/24

Link (Type-8) Link States (Area 0)

ADV Router	Age	Seq#	Link ID	Interface
1.1.1.1	579	0x80000001	5	Gi0/0/0
2.2.2.1	579	0x80000001	5	Gi0/0/0

Intra Area Prefix Link States (Area 0)

ADV Router	Age	Seq#	Link ID	Ref-lsttype	Ref-LSID
2.2.2.1	539	0x80000001	5120	0x2002	5
3.3.3.1	512	0x80000001	5120	0x2002	5

Lab - Implement Multiarea OSPFv3

Router Link States (Area 1)

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
1.1.1.1	602	0x80000001	0	0	None

OSPFv3 123 address-family ipv6 (router-id 1.1.1.1)

Router Link States (Area 0)

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
1.1.1.1	530	0x80000005	0	1	B
2.2.2.1	508	0x80000009	0	2	None
3.3.3.1	508	0x80000006	0	1	B

Net Link States (Area 0)

ADV Router	Age	Seq#	Link ID	Rtr count
2.2.2.1	539	0x80000001	5	2
3.3.3.1	511	0x80000001	5	2

Inter Area Prefix Link States (Area 0)

ADV Router	Age	Seq#	Prefix
1.1.1.1	579	0x80000001	2001:DB8:ACAD:1001::/64
1.1.1.1	559	0x80000001	2001:DB8:ACAD:1002::/64
3.3.3.1	551	0x80000001	2001:DB8:ACAD:2001::/64
3.3.3.1	512	0x80000001	2001:DB8:ACAD:2002::/64

Link (Type-8) Link States (Area 0)

ADV Router	Age	Seq#	Link ID	Interface
1.1.1.1	578	0x80000002	5	Gi0/0/0
2.2.2.1	578	0x80000002	5	Gi0/0/0

Intra Area Prefix Link States (Area 0)

ADV Router	Age	Seq#	Link ID	Ref-lstype	Ref-LSID
2.2.2.1	539	0x80000001	5120	0x2002	5
3.3.3.1	511	0x80000001	5120	0x2002	5

Router Link States (Area 1)

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
1.1.1.1	553	0x80000006	0	1	B
1.1.1.2	552	0x80000025	0	1	None

Net Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Rtr count
------------	-----	------	---------	-----------

```
1.1.1.2          560          0x80000001  38          2
```

Inter Area Prefix Link States (Area 1)

ADV Router	Age	Seq#	Prefix
1.1.1.1	578	0x80000001	2001:DB8:ACAD:A001::/64
1.1.1.1	538	0x80000001	2001:DB8:ACAD:A002::/64
1.1.1.1	506	0x80000001	2001:DB8:ACAD:2002::/64
1.1.1.1	506	0x80000001	2001:DB8:ACAD:2001::/64

Link (Type-8) Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Interface
1.1.1.1	559	0x8000000C	6	Gi0/0/1
1.1.1.2	598	0x80000002	38	Gi0/0/1

Intra Area Prefix Link States (Area 1)

ADV Router	Age	Seq#	Link ID	Ref-lsttype	Ref-LSID
1.1.1.2	481	0x80000016	0	0x2001	0
1.1.1.2	560	0x80000001	38912	0x2002	38

Part 5: Tune OSPFv3

Step 1: Configuring a passive interface.

- a. To configure a passive interface in traditional OSPFv3, use the **passive-interface** command in OSPFv3 router mode.

```
D1(config)# ipv6 router ospf 123
D1(config-rtr)# passive-interface e0/1
```

- b. To configure a passive interface in OSPFv3 with AF, you can use the **passive-interface** command in OSPFv3 router mode to configure the passive interface for both IPv4 and IPv6 AFs.

```
D2(config)# router ospfv3 123
D2(config-router)# passive-interface e0/1
```

- c. As an alternative, you can use the **passive-interface** command within AF configuration mode to configure the passive interface for a specific AFs.

```
D2(config-router)# no passive-interface e0/1
D2(config-router)# address-family ipv4 unicast
D2(config-router-af)# passive-interface e0/1
D2(config-router-af)# exit-address-family
D2(config-router)# address-family ipv6 unicast
D2(config-router-af)# passive-interface e0/1
D2(config-router-af)# exit-address-family
```

Step 2: Configuring summarization.

- a. The **area range ipv6-summary-address** command is used to summarize prefixes from one area into another. The *area* is the area from which the prefixes are summarized.


```
R1(config)# router ospfv3 123
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)# area 1 range 2001:db8:acad:1000::/52
```

```
R3(config)# router ospfv3 123
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)# area 2 range 2001:db8:acad:2000::/52
```

- b. Notice that R2 is now receiving the summarized prefixes.

```
R2# show ipv6 route ospf
<output omitted>
OI 2001:DB8:ACAD:1000::/52 [110/12]
   via FE80::1:2, GigabitEthernet0/0
OI 2001:DB8:ACAD:2000::/52 [110/12]
   via FE80::3:2, GigabitEthernet0/1
```

Why is prefix summarization considered desirable? How does it stabilize routing?

Step 3: Modifying the network type.

- a. OSPFv3 supports the same network types as OSPFv2. Notice that the Ethernet interfaces between R2 and R1, and R2 and R3, elect a DR and a BDR. This is because Ethernet is a multiaccess network. However, these are point-to-point links and there is no need for a DR or BDR.

```
R2# show ospfv3 interface brief
```

Interface	PID	Area	AF	Cost	State	Nbrs	F/C
Gi0/1	123	0	ipv4	1	DR	1/1	
Gi0/0	123	0	ipv4	1	DR	1/1	
Gi0/1	123	0	ipv6	1	DR	1/1	
Gi0/0	123	0	ipv6	1	DR	1/1	

- b. These connections can be changed to point-to-point using the **ospfv3 network point-to-point** interface command. This command needs to be configured on both sides of the point-to-point interface.

```
R2(config)# interface G0/1
R2(config-if)# ospfv3 network point-to-point
R2(config-if)# exit
R2(config)# interface G0/0
R2(config-if)# ospfv3 network point-to-point
```

```
R1(config)# interface G0/0
R1(config-if)# ospfv3 network point-to-point
```

```
R3(config)# interface G0/0
R3(config-if)# ospfv3 network point-to-point
```

- c. Notice that the links have now changed to P2P.

```
R2# show ospfv3 interface brief
```

Interface	PID	Area	AF	Cost	State	Nbrs	F/C
Gi0/1	123	0	ipv4	1	P2P	1/1	
Gi0/0	123	0	ipv4	1	P2P	1/1	
Gi0/1	123	0	ipv6	1	P2P	1/1	
Gi0/0	123	0	ipv6	1	P2P	1/1	

What is the effect on the state of the interface when changing a broadcast network to point-to-point?

Step 4: Advertising a default route.

- a. Similar to OSPFv2, an ASBR in OSPFv3 advertises using the **default-information** command. Configure a static default route for IPv4 and IPv6 on R2.

Note: Without a default route in the routing table, OSPF would require the **default-information originate always** command to advertise a default route.

```
R2(config)# ipv6 route ::/0 lo0
R2(config)# ip route 0.0.0.0 0.0.0.0 lo0

R2(config)# router ospfv3 123
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)# default-information originate
R2(config-router-af)# exit
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)# default-information originate
R2(config-router-af)# exit
```

- b. Verify D1 is receiving an IPv6 default route via OSPFv3.

```
D1# show ipv6 route ospf
<output omitted>
OE2 ::/0 [110/1], tag 123
    via FE80::1:1, Ethernet0/0
OI  2001:DB8:ACAD:2000::/52 [110/23]
    via FE80::1:1, Ethernet0/0
OI  2001:DB8:ACAD:A001::/64 [110/11]
    via FE80::1:1, Ethernet0/0
OI  2001:DB8:ACAD:A002::/64 [110/12]
    via FE80::1:1, Ethernet0/0
```

- c. Verify D2 is receiving an IPv4 default route via OSPFv3.

```
D2# show ip route ospfv3
<output omitted>
Gateway of last resort is 10.10.4.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.10.4.1, 00:01:04, Ethernet0/0
    10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
O IA   10.10.0.0/24 [110/13] via 10.10.4.1, 00:02:42, Ethernet0/0
    172.16.0.0/30 is subnetted, 2 subnets
O IA   172.16.0.0 [110/12] via 10.10.4.1, 00:02:42, Ethernet0/0
O IA   172.16.1.0 [110/11] via 10.10.4.1, 00:17:40, Ethernet0/0
```